

JOEL NIRMAL OOMMEN

Software Engineering Graduate | Machine Learning & Full-Stack Development

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PROFESSIONAL SUMMARY

Final-year Software Engineering student at the University of Stirling with hands-on experience in machine learning, NLP, and full-stack development. For honors dissertation a semantic search and classification system for news articles using scikit-learn, TF-IDF, and Hugging Face Transformers was built. Two internships across corporate IT and autonomous robotics, strong academic project portfolio in Java, Python, and Kotlin, and a track record of delivering working systems against deadlines.

EDUCATION

BSc (Hons) Software Engineering

University of Stirling, RAK Campus

Sep 2022 - Jun 2026 (Expected)

Ras Al Khaimah, UAE

- Relevant coursework: Machine Learning, Distributed Systems, Web Services, Android Development, Computer Games, UX Design, Data Structures & Algorithms, Databases
- Active Module Representative - liaising between students and academic faculty across multiple semesters

Higher Secondary Certificate (CBSE, Class 12)

Global Indian School, Ajman, UAE

2022

Scored 80%

Secondary School Certificate (CBSE, Class 10)

Our Own English High School, Sharjah, UAE

2020

Scored 94%

TECHNICAL SKILLS

Programming Languages: Python, Java, Kotlin, JavaScript, SQL, C, OCaml, HTML5, CSS3

Machine Learning & Data: scikit-learn, Hugging Face Transformers, TF-IDF, Cosine Similarity, Logistic Regression, Random Forest, SVM, Orange, Power BI, Pandas

Frameworks & Tools: Streamlit, Spring Boot, Android SDK (Jetpack Compose), Node.js, Express, Git, GitHub

Databases: MySQL, Microsoft SQL Server, MongoDB, SQLite, SharedPreferences

Concepts: REST APIs, HATEOAS, JWT Authentication, Distributed Systems, Computer Vision, Agile/CRISP-DM, Responsive Web Design, WCAG 2.1 Accessibility

PROJECTS

Smart Document Management for News Articles (Honours Dissertation)

Python, scikit-learn, Hugging Face, Streamlit

Supervised by Dr. Rajesh Subramanian

- Building a web-based document management system that ingests news articles and delivers semantic search, automated classification, and text summarization through a single Streamlit interface
- Implemented semantic search using TF-IDF vectorization with cosine similarity, enabling context-aware retrieval across the BBC News and UCI News Aggregator datasets
- Trained multi-class news classifiers with scikit-learn to auto-tag articles by category, and integrated Hugging Face Transformer models for abstractive summarization of long-form content
- Completed full ethics approval documentation including participant information sheets, consent forms, and structured user testing protocols

Store Profitability Classifier - Supervised ML

Python, scikit-learn, Orange

- Built and compared three classifiers (Logistic Regression, Random Forest, SVM) to predict retail store profitability from 19 features across 132 store records, following CRISP-DM methodology end-to-end
- Achieved 83% classification accuracy and 89.1% AUC with L1-regularized Logistic Regression (C=1) after 5-fold cross-validation and systematic hyperparameter tuning
- Executed full data-preparation pipeline: removed anomalous staff counts, merged inconsistent categorical values, standardized features, and reduced feature space by dropping low-signal variables

RESTful Drop-Shipping Order API

Java, Spring Boot, JWT, HATEOAS

- Designed and built a REST API for a drop-shipping retailer handling product browsing, order lifecycle, and operator management, with JWT-based authentication protecting privileged endpoints
- Implemented dynamic wholesaler integration by navigating an external HATEOAS API to auto-discover products, sync stock, and apply a 30% retail markup with real-time price validation at order time
- Applied HATEOAS throughout all resource responses with conditional state-based links (e.g. cancel link only for pending orders), making the API self-documenting

Distributed Mutual Exclusion (DME) System

Java, Socket Programming, Multithreading

- Implemented a centralized token-passing mutual exclusion system over TCP sockets coordinating multiple concurrent nodes, with a thread-safe priority buffer supporting three priority levels
- Added anti-starvation logic that auto-promotes low-priority requests after a 10-second wait threshold, preventing indefinite blocking while preserving priority scheduling
- Built coordinator fault tolerance: nodes detect failures via SocketTimeoutException, retry up to 3 times before graceful termination, and a dedicated shutdown protocol closes resources cleanly

Autonomous Object Detection & Sorting Robot

Python, OpenCV, Robotics

- Contributed to a robot that autonomously identifies objects by color and sorts them using computer vision and coordinated path planning
- Set up camera calibration and implemented color-based object recognition through image processing techniques to ensure reliable detection across varying lighting conditions
- Integrated robotic arm control with vision and navigation subsystems so the robot could locate, pick, and place objects with reliable end-to-end coordination

Diary Entry Android App

Kotlin, Android SDK, Fragments

- Built a single-Activity, multi-Fragment Android app with ViewPager2 and TabLayout navigation, supporting date selection, entry creation, and entry viewing across three dedicated screens
- Added features beyond the assignment brief including persistent storage via SharedPreferences with Gson, photo attachments, full-text search, and entry management (edit, delete, pin-as-favorite)
- Applied separation of concerns by centralizing data operations in a dedicated storage class, making fragments independent of persistence details

EXPERIENCE

IT Intern -Autonomous Robotics Project

Lab of Future

May 2025 - Jul 2025

UAE

- Contributed to the end-to-end development of an autonomous object detection and sorting robot, covering camera calibration, color-detection algorithms, path planning, and robotic arm integration
- Participated in robotic arm programming and integration, ensuring reliable coordination between the vision, navigation, and mechanical subsystems

- Conducted system testing, troubleshooting, and iterative optimization to improve reliability and reduce failure cases during repeated sort operations

Summer Intern -Motor Claims Department

Jun 2024 - Sep 2024

The Oriental Insurance Co. Ltd

UAE

- Provided technical support for mission-critical claims software, including deployment, configuration, troubleshooting, and routine maintenance to keep systems stable across daily operations
- Coordinated workflow between Motor Surveyors and the Claims Department, ensuring claim files progressed through processing and tracking without delay
- Managed claim documentation with attention to accuracy and completeness, supporting timely claim resolution for customers

CERTIFICATIONS & TRAINING

- AI Fluency: Framework & Foundations - Anthropic
- McKinsey Forward Program - McKinsey.org
- Robotics and AI - Great Learning
- Ultimate Java Part 1: Fundamentals - Code with Mosh
- Power BI Essential Training - LinkedIn Learning
- Introduction to Databases and SQL Querying - Udemy
- 2D Game Design and Development Essential Training - LinkedIn Learning
- One Million Prompts - H.H. Sheikh Hamdan bin Mohammed bin Rashid Al Maktoum
- My Stirling Award - University of Stirling

LANGUAGES

English (Fluent) | Malayalam (Fluent) | Hindi (Advanced) | Arabic (Basic)

ADDITIONAL INFORMATION

- UAE driving license holder